

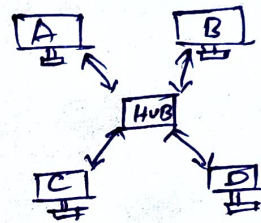
CCNA Network Devices

Behaviour of Hub :

Pros

1. It is a layer 1 device
2. It is used to connect multiple devices
3. It helps to broadcast and does not learn any address.

Cons



1. If a hub is connected to multiple devices like A, B, C, D and data is transferred from two or more devices simultaneously there will be loss in data transfer and that loss is known as Collision.
2. And due to broadcasting every time there will be high traffic in network.
3. When two or more devices transmit data at same exact time the data packets will collide and forcing both the devices to stop wait and resend the information.

SWITCH

* Behaviour:-

1. It is a Layer 2 device and connect multiple devices.
2. It helps in broadcast, unicast and multicast a data.
3. It will maintain a MAC Table
4. Also do a main function called address learning

When a data is entered in a switch, it immediately note down the incoming port number and source mac

5. Now data transfer:-

Once the data ~~transferred~~ received, then ^{the} switch checks the mac table, if there is ^{one} destination mac then it unicast, if multiple destination mac is there, then except the incoming port the data transmit to the destination port known as multicast.

6. If there is no D-MAC in the table then it broadcast.

Cons:-

Due to broadcasting, if there is multiple departments connected to various switch and those various switches are connected one to one, there will be huge traffic.

ROUTER:-

- It is a layer 3 device.
- It is also used to connect multiple devices.
- It is used to unicast and multicast data.
- It maintains a route table which contains routes.
- It helps in avoiding unnecessary broadcast, due to this network traffic will be less and network will be smooth and faster.

If Data transmitted to Router:-

• Once the router receives a data from a device, it finds whether there is route for destination IP in the routing table.

• If there is route then it unicasts the data.

• If a device transmits a data to router, router creates a routing table based on the network.

• It capture 3 major data:-

network ID

prefix

Interface name.